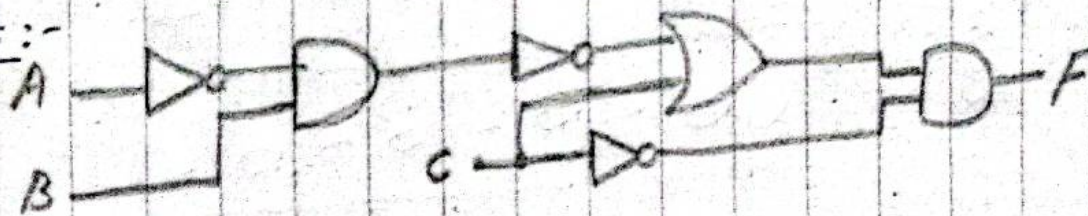


Boolean Expression: $((A'B)' + C) \cdot C'$

A	B	C	Not A	Not B	Not C	And(A'B)	$(A'B)' + C$	$((A'B)' + C) \cdot C'$	F
0	0	0	1	1	1	0	1	1	1
0	0	1	1	1	0	0	1	0	0
0	1	0	1	0	1	1	0	1	1
0	1	1	1	0	0	1	1	0	0
1	0	0	0	1	1	0	1	1	1
1	0	1	0	1	0	0	1	0	0
1	1	0	0	0	1	0	1	1	1
1	1	1	0	0	0	0	1	0	0

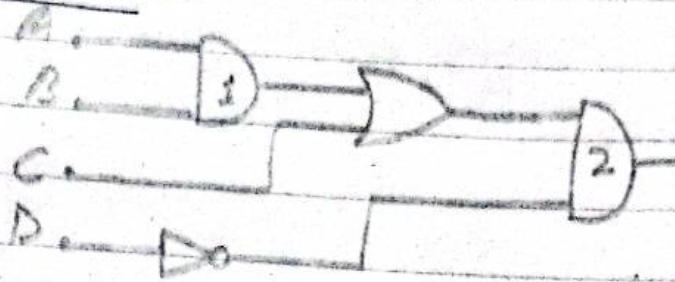
Circuit:-



Lab Task #1:-

Roll-no:- 24K-0737

Lab Task #3:-



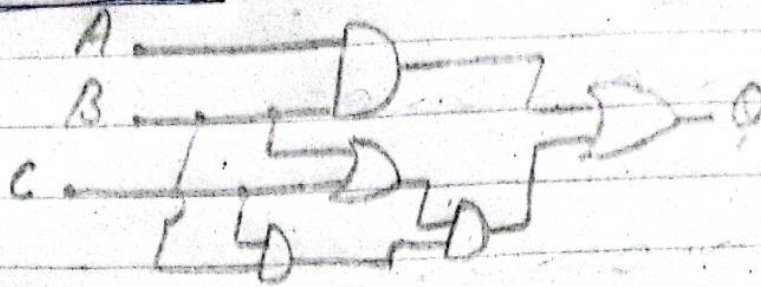
Boolean Expression:-

$$(AB + C) \cdot D'$$

Truth Table:-

A	B	C	D	AB	Not D	(AB+C)	(AB+C) · D'
0	0	0	0	0	1	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	1	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	1	0	1
0	1	1	1	0	0	1	0
1	0	0	0	0	1	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	1	1	1
1	0	1	1	0	0	1	0
1	1	0	0	1	1	1	1
1	1	0	1	1	0	1	0
1	1	1	0	1	1	1	1
1	1	1	1	1	0	1	0

LAB Task #4 :-



Boolean Expression:- $((B+C)(\overline{C}B) + AB)$

A	B	C	B+C	$(B+C)(\overline{C}B)$	AB	$(B+C)(\overline{C}B) + AB$
0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	1	0	1	0	0	0
0	1	1	1	1	0	1
1	0	0	0	0	0	0
1	0	1	1	0	0	0
1	1	0	1	0	1	1
1	1	1	1	1	1	1